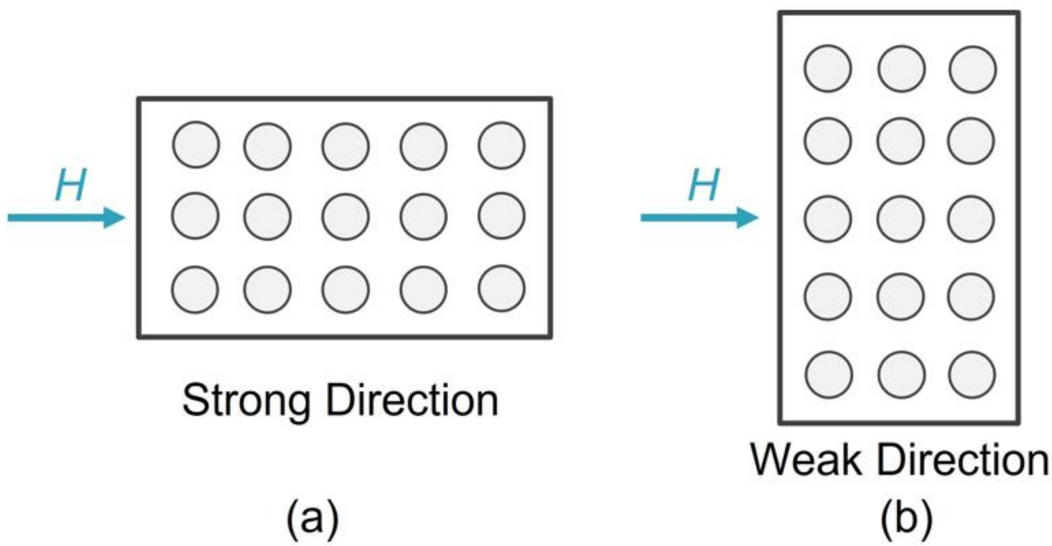
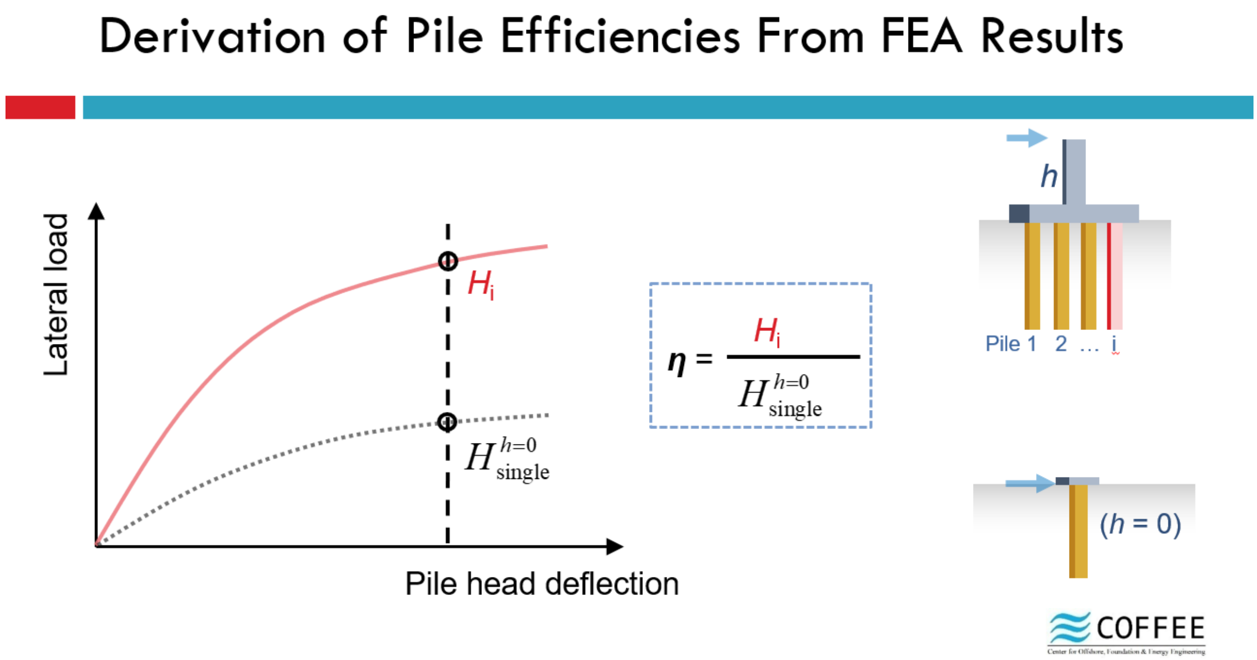
Problem Statement

The ultimate prediction goal is to predict the pile's lateral capacity when this specific pile is inside a pile group. We assume that we can obtain a pile's lateral capacity, whatever technique is available to us. So basically, we are trying to predict the group effect.

In this research project, we always assume that the lateral force the pile group is subjected to goes across the central line of the pile group. The term "pile head" always refers to the elevation at the soil surface. We use the term "load eccentricity" to describe the vertical distance between the loading point and the pile head. We assume that the single pile's lateral capacity is defined as the lateral force with a load eccentricity of 0m, under which the pile head will move 5cm laterally. The group piles' capacity are defined in the same way. The pile group's total lateral capacity is defined as the sum of the lateral capacity of all the piles.

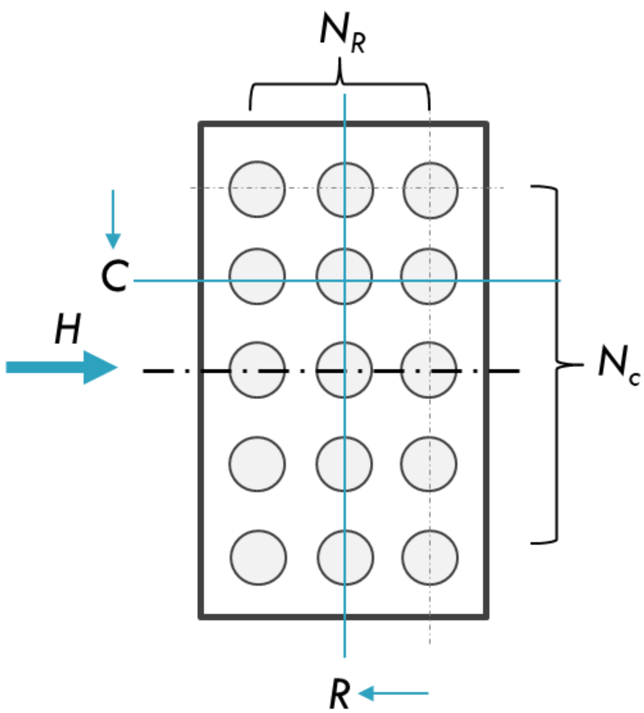




Input Parameters

The input parameters are summarized in the table below. The definition of “Rows” and “Columns” can be found in the picture below.

Because the whole database only contains sand scenario, relative density is the only parameter that is used here to capture the impact of soil property. Note that because of symmetry, the column number is counted from the nearest edge of the pile group.



Factors	Units	Meaning
h	m	Load eccentricity
R	1	The row number of the pile counting from the leading row
C	1	The column number of the pile counting from the nearest edge column of the pile group
N_c	1	Total number of columns in the pile group
N_R	1	Total number of rows in the pile group
S_p	1	The spacing between piles in the pile group, normalized by pile diameter
D_R	1	Relative density ($0 < D_R < 1$)

Output Parameter

The output parameter is a single value, which is pile capacity. The only restriction of the pile capacity is that it must be larger than zero. It is a float number, in 2/3 of the time, smaller than 1.

Quality of Database

As it always is in geotechnical problems, the database is sparse in some dimensions. For example, the relative density included in this database contains only 3 points, the pile spacing has only two choices. Within these, the spacing term is most problematic. As it is known, physical restrictions are less important when the database is dense enough. However, when spacing has only two choices, for extrapolation purposes, we have no other choice to involve our own observations or our empirical experience in our learning process.

Physical Restrictions, Limitations and Practical Restriction

1. The pile efficiency should increase monotonically with pile spacing. However, the impact of the spacing term should decrease when pile spacing is large enough. Which means it should have a limit. It usually does not hurt to assume this limit to be 1.0.
2. When the soil is very loose, then the group effect caused by the interaction between soils and piles should be smaller than it is for the dense sand. However, note that relative density equals to 0 doesn't mean there's no soil. It simply means that the sand is now on its loosest state.
3. When there's only one single pile, then C=1, R=1, Nc=1, Nr=1, then the pile capacity should only be dependent on load eccentricity, relative density should also go away from the equation. Therefore, we hope that the relative density term is multiplied by either row number or column number, and we hope that there is an independent term for load eccentricity h.
4. When load eccentricity increases, the pile efficiency must decrease. This is a rule of thumb. When load eccentricity increases to infinity, the piles cannot stand any lateral load. So, the pile efficiency should decrease to 0.
5. Practically speaking, we hope that geometry terms, load eccentricity terms, relative density terms can be separated. Spacing term is forced to be separated for reasons mentioned in the first item in this list.
6. The equation should not be very sensitive to the precision of the coefficients. We should be allowed to truncate the coefficients to 10^{-2}, and after truncation, some minor adjustments may be needed to make the equation make sense.